

St George Girls High School

Year 12

Mid-HSC Course Examination

2007



Mathematics Extension 1

General Instructions

- Working time – 1½ hours
- Reading time – 5 minutes
- Write using blue or black pen
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.
- Write on one side of the page only.
- Start each question in a **new booklet**.

Total marks – 60

- Attempt Questions 1 – 5
- All questions are of equal value

Question 1 (12 marks)

Marks

- a) The growth rate per hour, $\frac{dP}{dt}$, of a population of bacteria, P , is 12% of the population at that time. Initially the population is 100 000.

- (i) Show that the population in any hour can be calculated by the model:

1

$$P = P_0 e^{0.12t}$$

- (ii) Sketch the curve of population against time

1

- (iii) Determine the population after 4 hours

2

- b) An amount of water, W litres, in a tank, evaporates at a rate proportional to the amount of water in the tank at that time. This can be represented by $\frac{dW}{dt} \propto W$.

Initially the tank is full and the quantity is reduced by $\frac{1}{4}$ after 120 hours.

- (i) Show that this situation can be represented by $W = W_0 e^{kt}$

1

- (ii) Find an expression for the exact value of k

2

- (iii) What exact fraction of the water has evaporated after 240 hours?

1

- (iv) When will there be only $\frac{1}{4}$ of the water left in the tank?

3

Question 2 (12 marks)

Marks

- a) A cubic polynomial $P(x)$ gives remainders of 1 and 2 when divided by $x + 2$ and $x - 1$ respectively. Find the remainder when it is divided by $(x + 2)(x - 1)$.

Hint: let the remainder be linear.

3

- b) (i) Factorise $P(x) = x^4 - 2x^3 - 3x^2 + 8x - 4$ completely.

4

- (ii) Hence, sketch $y = x^4 - 2x^3 - 3x^2 + 8x - 4$ without using calculus, showing all intercepts.

2

- c) Solve $x^3 + 6x^2 + 11x + 6 = 0$ if the roots are in arithmetic progression.

3

Question 3 (12 marks)

Marks

a) If α, β, γ are the roots of the equation $x^3 - 2x + 5 = 0$, find:

(i) $\alpha + \beta + \gamma$

1

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma$

1

(iii) $\alpha\beta\gamma$

1

(iv) $\alpha^2 + \beta^2 + \gamma^2$

2

(v) $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$

1

(vi) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$

3

b) The line $y = 3x - 2$ is the tangent to the curve $y = x^3$ at the point $(1, 1)$. Find the point where this tangent meets the curve again.

Question 4 (12 marks)

Marks

- a) Solve $\cos 2x - 3\cos x = 1$, for $0^\circ \leq x \leq 360^\circ$ 4
- b) Find possible values of a if the lines $2x + 3y - 5 = 0$ and $ax + 2y + 3 = 0$ are inclined to each other at 45° 4
- c) Sketch the graph of $y = \sin(2x - \frac{\pi}{4})$ for $0 \leq x \leq \pi$ 2
- d) OAB is a sector of a circle, with radius 3cm and an angle of 30° subtended at the centre, O, of the circle. Find the **exact** area of the minor segment formed by the chord AB. 2

Question 5 (12 marks)

Marks

a) Differentiate:

(i) $\log_e(\cos x)$

1

(ii) $\sin^3(2x+1)$

2

b) Find the equation of the tangent at $x = \frac{\pi}{3}$ on the curve $y = \tan x$

3

c) (i) Evaluate $\int_0^{\frac{\pi}{6}} \sin 2x \, dx$

2

(ii) Find $\int \tan^2(x+1) \, dx$

2

d) Find the area of the region bounded by the curve $y = \sin x$ and the x -axis from $x = \frac{\pi}{2}$
to $x = -\frac{\pi}{2}$

2

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

SOLUTIONS

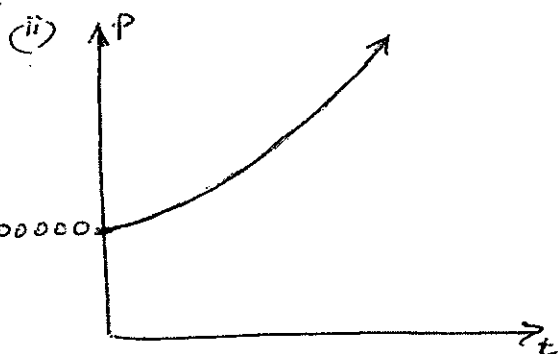
Question 1

a) i) $P = P_0 e^{0.12t}$

$$\frac{dP}{dt} = 0.12 P_0 e^{0.12t}$$

$$= 0.12 P$$

$\therefore \frac{dP}{dt}$ is 12% of the population at any time t .



(iii) $P = 100000 e^{0.12 \times 4}$

$$= 100000 e^{0.48}$$

$$\approx 161600$$

b) (i) $W = W_0 e^{kt}$

$$\frac{dW}{dt} = kW_0 e^{kt}$$

$$= kW$$

$\therefore \frac{dW}{dt} \propto W$

(ii) when $t = 120$

$$W = \frac{3}{4} W_0$$

$$\frac{3}{4} W_0 = W_0 e^{120k}$$

$$\frac{3}{4} = e^{120k}$$

$$120k = \log_e \frac{3}{4}$$

$$k = \frac{1}{120} \log_e \frac{3}{4}$$

(iii) $N = W_0 e^{\frac{1}{120} \log_e \frac{3}{4} \times 240}$

$$= W_0 e^{2 \log_e \frac{3}{4}}$$

$$= W_0 e^{\log_e \frac{9}{16}}$$

$$= W_0 \times \frac{9}{16}$$

$\therefore$ there is $\frac{9}{16}$ of the water left after 240 hours

$\therefore \frac{7}{16}$ of the water has evaporated

(iv) $\frac{1}{4} W_0 = W_0 e^{\frac{1}{120} \log_e \frac{3}{4} \times t}$

$$\frac{1}{4} = e^{\frac{t}{120} \log_e \frac{3}{4}}$$

$$\log_e \frac{1}{4} = \frac{t}{120} \log_e \frac{3}{4}$$

$$120 \log_e \frac{1}{4} = t \log_e \frac{3}{4}$$

$$t = \frac{120 \log_e \frac{1}{4}}{\log_e \frac{3}{4}}$$

$$= 578.26 \text{ hours}$$

Question 2

a) Let $R(x)$ be the remainder

$$R(x) = ax + b$$

$$R(-2) = 1$$

$$-2a + b = 1 \quad \text{--- (1)}$$

$$R(1) = 2$$

$$a + b = 2 \quad \text{--- (2)}$$

$$\text{(2) - (1)}$$

$$3a = 1$$

$$a = \frac{1}{3}$$

$$b = \frac{5}{3}$$

$\therefore$ the remainder will be

$$\frac{1}{3}x + \frac{5}{3}$$

b) (i) $P(1) = 0$

$$P(2) = 0$$

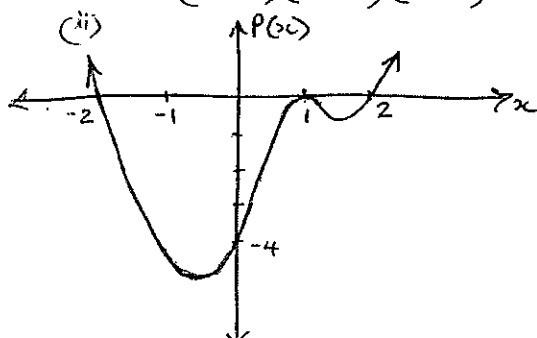
$$P(-2) = 0$$

$$\begin{array}{r} x^2 - 2x + 1 \\ x^2 - 4 \overline{) x^4 - 2x^3 - 3x^2 + 8x - 4} \\ \underline{x^4 \quad -4x^2} \\ -2x^3 + x^2 + 8x \\ \underline{-2x^3 + 8x} \\ x^2 -4 \\ \underline{x^2 -4} \\ 0 \end{array}$$

$$P(x) = (x^2 - 4)(x^2 - 2x + 1)$$

$$= (x^2 - 4)(x - 1)^2$$

$$= (x - 2)(x + 2)(x - 1)^2$$



c) Let the roots be α, β and $\frac{\alpha+\beta}{2}$

$$\alpha + \beta + \frac{\alpha+\beta}{2} = -6$$

$$2\alpha + 2\beta + \alpha + \beta = -12$$

$$3\alpha + 3\beta = -12$$

$$\alpha + \beta = -4 \quad \text{--- (1)}$$

$$\alpha\beta + \alpha\left(\frac{\alpha+\beta}{2}\right) + \beta\left(\frac{\alpha+\beta}{2}\right) = 11$$

$$\alpha\beta + \alpha \times -\frac{4}{2} + \beta \times -\frac{4}{2} = 11$$

$$\alpha\beta - 2\alpha - 2\beta = 11$$

$$\alpha\beta - 2(\alpha + \beta) = 11$$

$$\alpha\beta + 8 = 11$$

$$\alpha\beta = 3 \quad \text{--- (2)}$$

from (1)

$$\beta = -4 - \alpha \quad \text{--- (3)}$$

sub (3) into (2)

$$\alpha(-4 - \alpha) = 3$$

$$-4\alpha - \alpha^2 = 3$$

$$\alpha^2 + 4\alpha + 3 = 0$$

$$(\alpha + 1)(\alpha + 3) = 0$$

$$\therefore \alpha = -1 \quad \text{or} \quad \alpha = -3$$

$$\beta = -3 \quad \beta = -1$$

$\therefore$ the roots are $-3, -2, -1$

i.e. $x = -3, -2, -1$

is the solution

Question 3

a) (i) $\alpha + \beta + \gamma = 0$

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma = -2$

(iii) $\alpha\beta\gamma = -5$

(iv) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2$
 $-2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$$= 0^2 - 2 \times (-2)$$

$$= 4$$

(v) $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$

$$= (\alpha\beta\gamma)(\alpha + \beta + \gamma)$$

$$= -5 \times 0$$

$$= 0$$

(vi) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{(\alpha\beta\gamma)^2}$

$$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2$$

$$- 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2)$$

$$= (-2)^2 - 2 \times 0$$

$$= 4$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{4}{(-5)^2}$$

$$= \frac{4}{25}$$

b) $y = 3x - 2 \quad \text{--- (1)}$

$y = x^3 \quad \text{--- (2)}$

sub. (2) into (1)

$$x^3 = 3x - 2$$

$$x^3 - 3x + 2 = 0$$

$(x-1)^2$ is a solution to this equation

as $y = 3x - 2$ is a tangent to $y = x^3$ at $x = 1$

$$\begin{array}{r} x^3 - 3x + 2 \\ x^2 - 2x + 1 \overline{) x^3 - 3x + 2} \\ \underline{x^3 - 2x^2 + x} \\ 2x^2 - 4x + 2 \\ \underline{2x^2 - 4x + 2} \\ 0 \end{array}$$

$$\therefore (x-1)^2(x+2) = 0$$

at $x = -2 \quad y = -8$

$\therefore$ the tangent meets the curve again at $(-2, -8)$

Question 4

a) $\cos 2x - 3\cos x = 1$

$$2\cos^2 x - 1 - 3\cos x = 1$$

$$2\cos^2 x - 3\cos x - 2 = 0$$

$$(2\cos x + 1)(\cos x - 2) = 0$$

either

$$2\cos x + 1 = 0$$

$$\cos x = -\frac{1}{2}$$

$$x = 120^\circ, 240^\circ$$

or $\cos x - 2 = 0$

$$\cos x = 2$$

no solution

b) $m_1 = -\frac{2}{3} \quad m_2 = -\frac{a}{2}$

$$\left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \tan 45^\circ$$

$$\left| \frac{-\frac{2}{3} + \frac{a}{2}}{1 + \frac{a}{3}} \right| = 1$$

either $\frac{-\frac{2}{3} + \frac{a}{2}}{1 + \frac{a}{3}} = 1$

$$-\frac{2}{3} + \frac{a}{2} = 1 + \frac{a}{3}$$

$$\frac{a}{6} = \frac{5}{3}$$

$$a = 10$$

or $\frac{-\frac{2}{3} + \frac{a}{2}}{1 + \frac{a}{3}} = -1$

$$-\frac{2}{3} + \frac{a}{2} = -1 - \frac{a}{3}$$

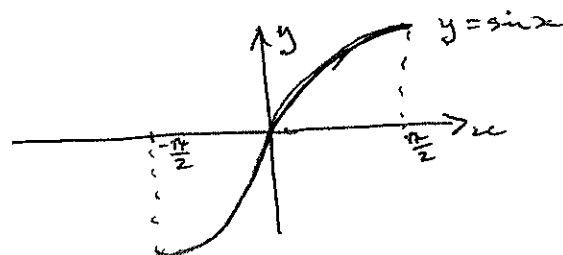
$$\frac{5a}{6} = -\frac{1}{3}$$

$$a = -\frac{6}{15}$$

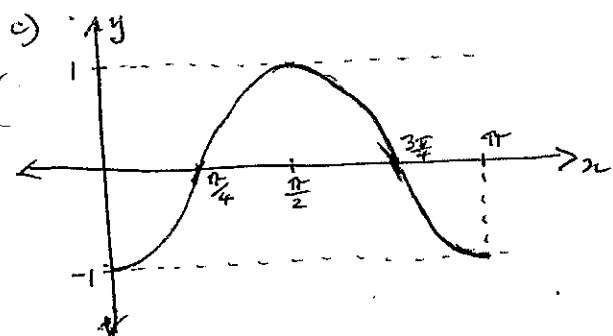
$$\begin{aligned} \text{c) (i)} \int_0^{\frac{\pi}{6}} \sin 2x \, dx &= \left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}} \\ &= -\frac{1}{2} \cos \frac{\pi}{3} - \left(-\frac{1}{2} \cos 0 \right) \\ &= -\frac{1}{4} - \left(-\frac{1}{2} \right) \\ &= -\frac{1}{4} + \frac{1}{2} \\ &= \frac{1}{4} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \int \tan^2(x+1) \, dx &= \int (\sec^2(x+1) - 1) \, dx \\ &= \int \sec^2(x+1) \, dx - \int 1 \, dx \\ &= \tan(x+1) - x + C \end{aligned}$$

d)



$$\begin{aligned} \text{Area} &= 2 \int_0^{\frac{\pi}{2}} \sin x \, dx \\ &= 2 \left[-\cos x \right]_0^{\frac{\pi}{2}} \\ &= 2(-0 + 1) \\ &= 2 \text{ units}^2 \end{aligned}$$



d)

$$\begin{aligned} \text{Area} &= \frac{1}{2} r^2 \theta - \frac{1}{2} r^2 \sin \theta \\ \theta &= \frac{\pi}{6}, r = 3 \\ \text{Area} &= \frac{9\pi}{12} - \frac{9}{2} \sin \frac{\pi}{6} \\ &= \frac{3\pi}{4} - \frac{9}{4} \\ &= \frac{3\pi - 9}{4} \text{ cm}^2 \end{aligned}$$

Question 5

$$\begin{aligned} \text{a) (i)} \frac{d}{dx} (\log_e(\cos x)) &= \frac{-\sin x}{\cos x} \\ &= -\tan x \end{aligned}$$

$$\begin{aligned} \text{(ii)} \frac{d}{dx} (\sin^3(2x+1)) &= 6 \cos(2x+1) \sin^2(2x+1) \end{aligned}$$

$$\begin{aligned} \text{b) } y &= \tan x \quad \frac{dy}{dx} = \sec^2 x \\ &= \sqrt{3} \quad = 4 \text{ when } x = \frac{\pi}{3} \\ \text{when } x &= \frac{\pi}{3} \\ y - \sqrt{3} &= 4(x - \frac{\pi}{3}) \end{aligned}$$

$\therefore$ the equation of the tangent is

$$4x - y + \sqrt{3} - \frac{4\pi}{3} = 0$$